

Assignment 14:

Deploy an Optimized ViT Model with LoRA, Quantization, FastAPI, and Docker

Objective

Build an end-to-end production-style image classification pipeline using a Vision Transformer (ViT).

You will:

- Fine-tune ViT using LoRA.
- Save the best LoRA model.
- Apply quantization to reduce the model's memory footprint.
- Build a FastAPI inference service.
- Containerize the inference application using Docker.
- Run inference through the Dockerized API.